

13.10 (单限电压比较器, 反馈回路稳压管限幅器)如图 13.10 所示单限电压比较器, 已知: 输入电压信号 $u_i = 5 \sin \omega t \text{V}$, 参考电压 $U_R = 2\text{V}$, 稳压管稳压值 $U_{Z1} = 5\text{V}$, $U_{Z2} = 3\text{V}$, 稳压管正向压降 $U_F = 0.7\text{V}$, 运放饱和输出电压 $U_{OM} = \pm 13\text{V}$ 。画出输出电压 u_o 的波形。

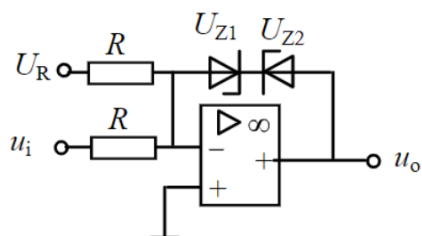
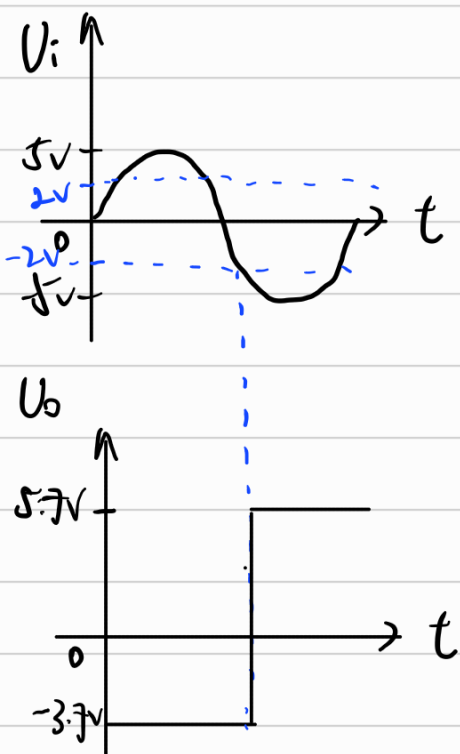


图 13.10



13.12 (方波发生器)图 13.12 所示电路, $R_f = 12 \text{ k}\Omega$, $R_F = 14 \text{ k}\Omega$, $R = 20 \text{ k}\Omega$,

$R_1 = 30 \text{ k}\Omega$, $C = 0.01 \mu\text{F}$, $U_{\text{OM}} = \pm 10 \text{ V}$ 。当电位器的滑动端分别置于图中 a、b、c 三个不同位置时 (b 为中点, a、c 为两 endpoint), 求输出电压 u_o 的频率 f , 并画出输出电压的波形。

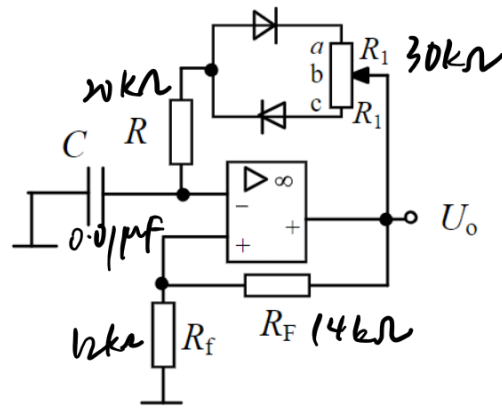


图 13.12

在 b 点位置时,

$$T_1 = T_2 \quad T = 2RC \ln\left(1 + \frac{2R}{R_F}\right)$$

$$T = 2(R + R_1)C \ln\left(1 + \frac{2R}{R_F}\right)$$

$$= 2(20 + 30) \times 10^3 \times 0.01 \times 10^{-6} \ln\left(1 + \frac{2(20)}{14}\right)$$

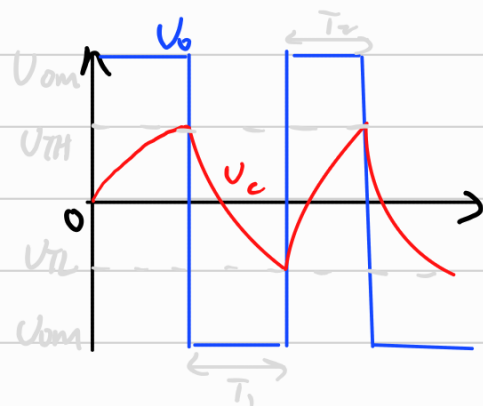
$$= 9.9853 \times 10^{-4} \text{ s} \approx 1 \text{ ms} \quad T_1 = T_2 = \frac{1}{2}T = 0.5 \text{ ms}$$

$$f = \frac{1}{T} \approx 1000 \text{ Hz}$$

$$U_{\text{TH}} = \frac{R_1}{R_1 + R_2} U_{\text{om}} \quad U_{\text{TL}} = -U_{\text{TH}}$$

$$= \frac{R_f}{R_f + R_F} U_{\text{om}}$$

$$= \frac{12}{26}(10) = 4.6154 \text{ V} \approx 4.62 \text{ V} \quad U_{\text{TL}} = -4.62 \text{ V}$$



在a点位置时,

$$T_1 = RC \ln\left(1 + \frac{2R}{R_F}\right)$$

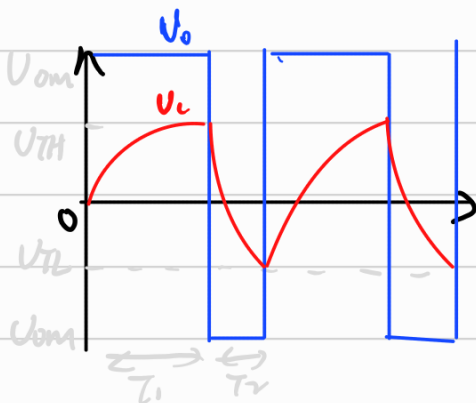
$$= 80 (0.01 \times 10^{-6}) \ln\left(1 + \frac{2(12)}{14}\right)$$

$$= 7.9882 \times 10^{-4} \approx 0.8 \text{ ms}$$

$$T_2 = 2 \times 10^3 (0.01 \times 10^{-6}) \ln\left(1 + \frac{2(12)}{14}\right)$$

$$\approx 0.2 \text{ ms}$$

$$T = T_1 + T_2 = 1 \text{ ms} \quad f = 1000 \text{ Hz}$$

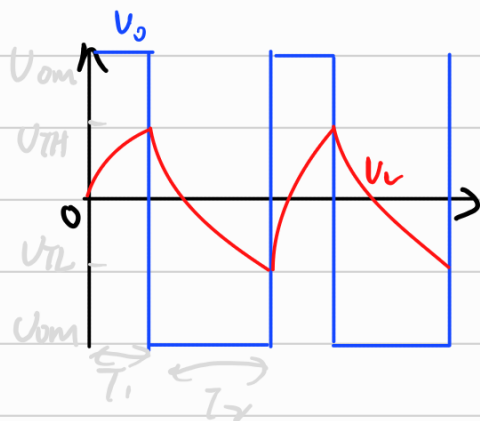


在c点位置:

刚好与a点情况相反

$$T_1 = 0.2 \text{ ms}$$

$$T_2 = 0.8 \text{ ms} \quad f \text{ 保持不变} = 1000 \text{ Hz}$$



13.14 (三角波、锯齿波发生器) 如图 13.14 三角波发生器, 求未接入 D 和 R_7 支路时三角波的频率。如果与 R_5 并联再接入 D 和 R_7 支路, 再计算三角波的频率。

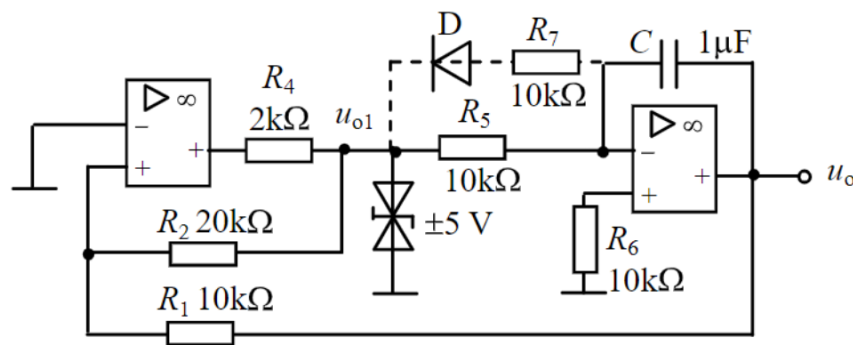


图 13.14

无 D 与 R_7

$$U_{TH} = \frac{R_1}{R_1 + R_2} U_Z = \frac{1}{2}(5) = 2.5V$$

$$U_{TL} = -2.5V$$

$$T = 4 \frac{R_1}{R_2} R_5 C = 4 \left(\frac{1}{2} \right) (10 \times 10^3) (1 \times 10^{-6})$$

$$= 2ms$$

$$f = \frac{1}{T} = 50Hz$$

接入 D 与 R_7

放电时,

$$U_{O1} = U_{om}$$

$$U_O = - \frac{U_{om}}{R_5 C} t$$

$$U_{TL} - U_{TH} = - \frac{2R_1}{R_2} U_{om} = - \frac{U_{om}}{R_5 C} T_1$$

$$T_1 = \frac{2R_1}{R_2} R_5 C = 0.01s$$

充电时,

$$U_{O1} = -U_{om}$$

$$U_O = \frac{U_{om}}{\frac{R_5}{2} C} t \quad U_{TH} - U_{TL} = \frac{2R_1}{R_2} U_{om} = \frac{2U_{om}}{R_5 C} t$$

$$T_2 = \frac{R_1}{R_2} R_5 C = \frac{1}{2} T_1 = 0.005s$$

$$T = T_1 + T_2 = 0.015s \quad f = \frac{1}{T} = 66.7Hz$$

13.15 (文氏桥正弦波发生器, 迟滞比较器) 图 13.15 所示电路, $R = 3.183 \text{ k}\Omega$, $C = 0.1 \mu\text{F}$, 运放 $U_{\text{OM}} = \pm 10 \text{ V}$ 。求电压 u_{o1} 的频率 f_1 , 并画出 u_{o1} 、 u_{o2} 的波形。

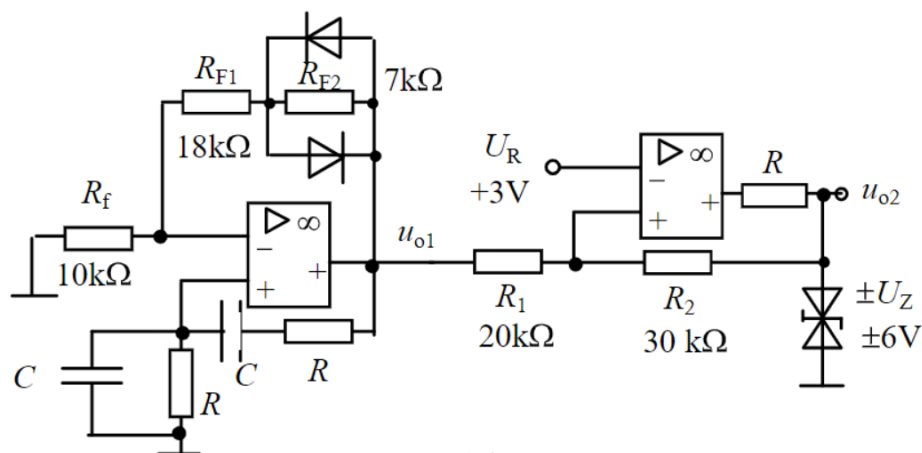


图 13.15

第一级

$$f_1 = \frac{1}{2\pi RC} = \frac{1}{2\pi (3.183 \times 10^3) (0.1 \times 10^{-6})} = 500 \text{ Hz}$$

第二级

$$\text{当 } u_1 < U_R, u_o = 6\text{V}$$

$$U_{TH} = \frac{R_1}{R_1 + R_2} U_Z + \frac{R_2}{R_1 + R_2} U_R = 9\text{V}$$

$$\text{当 } u_1 > U_R, u_o = -6\text{V}$$

$$U_{TL} = -\frac{R_1}{R_1 + R_2} U_Z + \frac{R_2}{R_1 + R_2} U_R = 1\text{V}$$

